Call for Papers of IEEE TCSVT Special Issue on

Towards Spatial Intelligence: 3D Understanding, Generation, and Reasoning from Real-World Videos

Aim and Scope

Recent advances in 3D vision and video understanding are enabling machines to reconstruct and reason about dynamic 3D environments from real-world videos. Together with large-scale video data and foundation models, these developments offer new opportunities to learn structured representations of geometry, appearance, motion, and semantics, advancing video-driven spatial intelligence.

Key challenges remain in maintaining temporal and geometric consistency under unconstrained camera motion, scaling to diverse in-the-wild data, integrating multimodal signals, and reasoning about spatial relationships, interactions, and scene dynamics.

This Special Issue invites research at the intersection of video understanding, 3D modeling and generation, and spatial reasoning, including novel representations, learning paradigms, foundation models, and real-world systems for video-driven spatial intelligence.

Topics of interest

- Advanced 3D compression (e.g., 3D Gaussian Splatting (3DGS) compression)
- Neural representations of (time-varying) 3D object/human/scene
- 3D human/object modeling, editing and animation
- 3D interaction modeling among human/object/scenes
- Video-to-3D generation with multimodal cues
- Human-in-the-loop 3D content generation
- Large-scale scene reconstruction/composition/generation
- Learning 3D world models from egocentric or in-the-wild video
- New benchmarks for 3D understanding/generation/reasoning
- Foundation models for 3D reconstruction/generation/reasoning
- Application enabled by spatial intelligence, e.g. AR/VR/Robotics

Important Dates

Submission deadline	October 31, 2026
First-round review	December 15, 2027
Revisions due	January 15, 2027
Second-round review	February 15, 2027
Final manuscripts	March 15, 2027
Publication date	March 31, 2027

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